CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

# Linear Transformations

Questions by topic 11 questions · 86 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

## Questions


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|---|---|---|
| Question 1 | *6 marks* | 2 |
| Question 2 | *10 marks* | 3 |
| Question 3 | *9 marks* | 4 |
| Question 4 | *10 marks* | 5 |
| Question 5 | *9 marks* | 6 |
| Question 6 | *8 marks* | 7 |
| Question 7 | *2 marks* | 8 |
| Question 8 | *5 marks* | 9 |
| Question 9 | *11 marks* | 10 |
| Question 10 | *4 marks* | 11 |
| Question 11 | *12 marks* | 12 |

**1.** The transformation $U$, represented by the $2 \times 2$ matrix $\mathbf{P}$, is a reflection in the $x$-axis.

**(a)** Write down the matrix $\mathbf{P}$. **[1]**

The transformation $V$, represented by the $2 \times 2$ matrix $\mathbf{Q}$, is a reflection in the line $y = x$.

**(b)** Write down the matrix $\mathbf{Q}$. **[1]**

Given that $U$ followed by $V$ is transformation $T$, which is represented by the matrix $\mathbf{R}$

**(c)** express $\mathbf{R}$ in terms of $\mathbf{P}$ and $\mathbf{Q}$, **[1]**

**(d)** find the matrix $\mathbf{R}$, **[1]**

**(e)** give a full geometrical description of $T$ as a single transformation. **[2]**

**2.** A right-angled triangle $T$ has vertices $A(-2,1)$, $B(1,1)$ and $C(1,5)$. When $T$ is transformed by the matrix

$$\mathbf{P} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

the image is $T'$.

**(a)** Find the coordinates of the vertices of $T'$. **[2]**

**(b)** Describe fully the transformation represented by $\mathbf{P}$. **[2]**

The matrices

$$\mathbf{Q} = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$$

and

$$\mathbf{R} = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}$$

represent two transformations. When $T$ is transformed by the matrix $\mathbf{QR}$, the image is $T''$.

**(c)** Find $\mathbf{QR}$. **[1]**

**(d)** Find the determinant of $\mathbf{QR}$. **[2]**

**(e)** Using your answer to part (d), find the area of $T''$. **[3]**

**3.** **(i)** In each of the following cases, find a $2 \times 2$ matrix that represents

(a) a reflection in the line $y = x$ **[1]**

(b) a rotation of $60°$ clockwise about $(0, 0)$ **[1]**

(c) a rotation of $60°$ clockwise about $(0, 0)$ followed by a reflection in the line $y = x$ **[1]**

**(ii)** The triangle $T$ has vertices at the points $(1, 2)$, $(5, 5)$ and $(k, -1)$, where $k > -3$ is a constant.

Triangle $T$ is transformed onto the triangle $T'$ by the matrix

$$\begin{pmatrix} 2 & 5 \\ 1 & -1 \end{pmatrix}$$

Given that the area of triangle $T'$ is 84 square units, find the value of $k$. **[6]**

**4.**

$$\mathbf{P} = \begin{pmatrix} p+1 & 2 \\ 2 & p+1 \end{pmatrix}$$

where $p$ is a positive constant.

The matrix $\mathbf{P}$ represents a linear transformation $U$.
The parallelogram $S$ has vertices at the points with coordinates $(1, 1)$, $(4, 2)$, $(6, 5)$ and $(3, 4)$.
The area of the image of $S$ under the linear transformation $U$ is 35.

**(a)** Determine the value of $p$. **[6]**

The transformation $V$ consists of a reflection in the line $y = x$ followed by a stretch of scale factor 2 parallel to the $x$-axis with the $y$-axis invariant. The transformation $V$ is represented by the matrix $\mathbf{Q}$.

**(b)** Write down the matrix $\mathbf{Q}$. **[3]**

Given that $U$ followed by $V$ is the transformation $W$, which is represented by the matrix $\mathbf{R}$

**(c)** find the matrix $\mathbf{R}$. **[1]**

**5.** Consider the matrix

$$\mathbf{A} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

**(i)** Calculate $\mathbf{A}^2$. **[1]**

You are told that the matrix $\mathbf{A}$ represents a reflection in a line through the origin.

**(ii)** Explain why your answer to part (i) is consistent with this information. **[1]**

**(iii)** By finding the invariant points of the transformation, determine the equation of the mirror line. **[2]**

**(iv)** Describe fully the transformation represented by

$$\mathbf{B} = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$ **[2]**

**(v)** A composite transformation is formed by the transformation represented by $\mathbf{A}$ followed by the transformation represented by $\mathbf{B}$. Find the single matrix that represents this composite transformation. **[1]**

**(vi)** The composite transformation described in part (v) is equivalent to a single reflection. Find the equation of the mirror line of this reflection. **[2]**

**6.** The matrix $\mathbf{M}$ is given by

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$$

where $k > 0$ and $k \neq 1$.

**(a)** The matrix $\mathbf{M}$ represents a sequence of two geometrical transformations. State the type of each transformation, and make clear the order in which they are applied. **[2]**

The unit square in the $x$-$y$ plane is transformed by $\mathbf{M}$ onto rectangle $OABC$.

**(b)** Find, in terms of $k$, the area of rectangle $OABC$ and the matrix which transforms $OABC$ onto the unit square. **[3]**

**(c)** Show that the line through the origin with gradient $\frac{1}{\sqrt{k}}$ is invariant under the transformation in the $x$-$y$ plane represented by $\mathbf{M}$. **[3]**

**7.** Bilal has been asked to describe the transformation given by the matrix

$$\begin{pmatrix} \frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \end{pmatrix}$$

He writes his answer as follows:

> The transformation is a rotation about the $y$-axis through an angle of $\theta$, where
>
> $$\cos\theta = \frac{\sqrt{3}}{2} \quad \text{and} \quad \sin\theta = \frac{1}{2}$$
>
> $$\theta = 30^\circ$$

Identify and correct the error in Bilal's work. **[2]**

**8.** You are given the matrix $\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$.

**(a)** Find $\mathbf{A}^5$. **[1]**

**(b)** Describe the transformation that $\mathbf{A}$ represents. **[2]**

The matrix $\mathbf{B}$ represents a reflection in the plane $x = y$.

**(c)** Write down the matrix $\mathbf{B}$. **[1]**

The point $Q$ has coordinates $(4, -1, 2)$. The point $Q'$ is the image of $Q$ under the transformation represented by $\mathbf{B}$.

**(d)** Find the coordinates of $Q'$. **[1]**

**9.** Given that the matrix $\mathbf{M}$, where

$$\mathbf{M} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

represents a rotation through $\alpha^\circ$ about the $y$-axis in the positive sense given by the right-hand rule

**(a)** determine the value of $\alpha$. **[1]**

**(b)** Hence determine the smallest possible positive integer value of $k$ for which $\mathbf{M}^k = \mathbf{I}$. **[2]**

The $3 \times 3$ matrix $\mathbf{N}$ represents a reflection in the plane with equation $x = 0$.

**(c)** Write down the matrix $\mathbf{N}$. **[1]**

The point $A$ has coordinates $(1, 2, 3)$.

The point $B$ is the image of the point $A$ under the transformation represented by matrix $\mathbf{M}$ followed by the transformation represented by matrix $\mathbf{N}$.

**(d)** Show that the coordinates of $B$ are $(2\sqrt{2}, 2, -\sqrt{2})$. **[2]**

Given that $O$ is the origin

**(e)** show that, to 3 significant figures, the size of angle $AOB$ is $79.4^\circ$. **[3]**

**(f)** Hence determine the area of triangle $AOB$, giving your answer to 3 significant figures. **[2]**

**10.** Three non-singular $3 \times 3$ matrices, $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{R}$, are such that

$$\mathbf{RA} = \mathbf{B}$$

The matrix $\mathbf{R}$ represents a rotation about the $x$-axis through an angle $\theta$ and

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\sin\theta & \cos\theta \\ 0 & \cos\theta & \sin\theta \end{pmatrix}$$

**(a)** Show that $\mathbf{A}$ is independent of the value of $\theta$. **[3]**

**(b)** Give a full description of the single transformation represented by the matrix $\mathbf{A}$. **[1]**

**11.** **(a)** Specify fully the transformation $T$ of the plane associated with the matrix $\mathbf{M}$, where

$$\mathbf{M} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

and $0 < \theta < 2\pi$. **[2]**

**(b)** (i) Find $\det(\mathbf{M})$. **[1]**

(ii) Deduce **two properties** of the transformation $T$ from the value of $\det(\mathbf{M})$. **[2]**

**(c)** Prove that

$$\mathbf{M}^n = \begin{pmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{pmatrix}$$

where $n$ is a positive integer. **[6]**

**(d)** Hence specify fully a **single** transformation which is equivalent to $n$ applications of the transformation $T$. **[1]**